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PHY 103 QUESTIONS AND ANSWER

1. Fluid is a substance that (a) cannot be subjected to shear forces (b) always expands until it fills any container (c) has the same shear stress.at a point regardless of its motion (d) cannot remain at rest under action of any shear force (e) flows. Ans: d

2. Fluid is a substance which offers no resistance to change of

(a) Pressure (b) flow (c) shape (d) volume (e) temperature. Ans: c

3. Practical fluids (a) are viscous (b) possess surface tension (c) are compressible (d) possess all the above properties (e) possess none of the above properties. Ans: d

4. in a static fluid (a) resistance to shear stress is small (b) fluid pressure is zero (c) linear deformation is small (d) only normal stresses can exist(e) viscosity is nil. Ans: d

5. A fluid is said to be ideal, if it is (a) incompressible (b) inviscous (c) viscous and incompressible (d) inviscous and compressible (e) inviscous and incompressible. Ans: e

6. An ideal flow of any fluid must fulfill the following (a) Newton's law of motion (b) Newton's law of viscosity (c) Pascal' law (d) Continuity equation (e) Boundary layer theory. Ans: d

7. If no resistance is encountered by displacement, such a substance is known as (a) fluid (b) water (c) gas (d) perfect solid (e) ideal fluid. Ans: e

8. The volumetric change of the fluid caused by a resistance is known as (a) volumetric strain (b) volumetric index (c) compressibility (d) adhesion (e) cohesion. Ans: c

9. Liquids (a) cannot be compressed (b) occupy definite volume (c) are not affected by change in pressure and temperature d) GO are not viscous (e) none of the above. Ans: e

10. Density of water is maximum at (a) 0°C (b) 0°K (c) 4°C (d) 100°C (e) 20°C. Ans: c

12. The value of mass density in kgsecVm4 for water at 0°C is (a) 1 (b) 1000 (c) 100 (d) 101.9 (e) 91

Ans: d

14. Property of a fluid by which its own molecules are attracted is called (a) adhesion (b) cohesion (c) viscosity (d) compressibility (e) surface tension. Ans: b

15. Mercury does not wet glass. This is due to property of liquid known as (a) adhesion (b) cohesion (c) surface tension (d) viscosity (e) compressibility. Ans: c

16. The property of a fluid which enables it to resist tensile stress is known as (a) compressibility (b) surface tension (c) cohesion (d) adhesion (e) viscosity. Ans: c

17. Property of a fluid by which molecules of different kinds of fluids are attracted to each other is called (a) adhesion (b) cohesion (c) viscosity (d) compressibility (e) surface tension. Ans: a

18. The specific weight of water is 1000 kg/m" (a) at norma] pressure of 760 mm (b) at 4°C temperature (c) at mean sea level (d) all the above (e) none of the above. Ans: d

19. Specific weight of water in S.I. units is equal to (a) 1000 N/m3 (b) 10000 N/m3 (c) 9.81 xlO3 N/m3 (d) 9.81 xlO6N/m3 (e) 9.81 N/m3. Ans: c

20. When the flow parameters at any given instant remain same at every point, then flow is said to be (a) quasi static (b) steady state (c) laminar (d) uniform (e) static. Ans: d

21. Which of the following is dimensionless (a) specific weight (b) specific volume (c) specific speed (d) specific gravity (e) specific viscosity. Ans: d

22. The normal stress in a fluid will be constant in all directions at a point only if (a) it is incompressible (b) it has uniform viscosity (c) it has zero viscosity (d) it is frictionless (e) it is at rest. Ans: e

23. The pressure at a point in a fluid will not be same in all the directions when the fluid is (a) moving (b) viscous (c) viscous and static (d) in viscous and moving (e) viscous and moving. Ans: e

24. An object having 10 kg mass weighs 9.81kg on a spring balance. The value of 'g' at this place is (a) 10m/sec2 (b) 9.81 m/sec2 (c) 10.2/m sec (d) 9.75 m/sec2 (e) 9 m/sec . Ans: a

25. The tendency of a liquid surface to contract is due to the following property (a) cohesion (b) adhesion (c) viscosity (d) surface tension (e) elasticity. Ans: d

26. The surface tension of mercury at normal temperature compared to that of water is (a) more (b) less (c) same (d) more or less depending on size of glass tube (e) none of the above. Ans: a

27. A perfect gas (a) has constant viscosity (b) has zero viscosity (c) is incompressible (d) is of theoretical interest (e) none of the above. Ans: e

32. For very great pressures, viscosity of moss gases and liquids (a) remains same (b) increases (c) decreases (d) shows erratic behaviour (e) none of the above. Ans: d

33. A fluid in equilibrium can't sustain (a) tensile stress (b) compressive stress (c) shear stress (d) bending stress (e) all of the above. Ans: c

34. Viscosity of water in comparison to mercury is (a) higher (b) lower (c) same (d) higher/lower depending on temperature (e) unpredictable. Ans: a

35. The bulk modulus of elasticity with increase in pressure (a) increases (b) decreases (c) remains constant (d) increases first up to certain limit and then decreases (e) unpredictable. Ans: a

36. The bulk modulus of elasticity (a) has the dimensions of 1/pressure (b) increases with pressure (c) is large when fluid is more compressible (d) is independent of pressure and viscosity (e) is directly proportional to flow. Ans: b

37. A balloon lifting in air follows the following principle (a) law of gravitation (b) Archimedes principle (c) principle of buoyancy (d) all of the above (e) continuity equation. Ans: d

38. The value of the coefficient of compressibility for water at ordinary pressure and temperature in kg/cm is equal to (a) 1000 (b) 2100 (c) 2700 (d) 10,000 (e) 21,000. Ans: e

39. The increase of temperature results in (a) increase in viscosity of gas (b) increase in viscosity of liquid (c) decrease in viscosity of gas (d) decrease in viscosity of liquid (e) (a) and (d) above. Ans: d

40. Surface tension has the units of (a) newtons/m (b) newton's/m (c) newton/m (d) newtons (e) newton m. Ans: c

41. Surface tension (a) acts in the plane of the interface normal to any line in the surface (b) is also known as capillarity (c) is a function of the curvature of the interface (d) decreases with fall in temperature (e) has no units. Ans: a

42. The stress-strain relation of the newtoneon fluid is (a) linear (b) parabolic (c) hyperbolic (d) inverse type (e) none of the above. Ans: a

43. A liquid compressed in cylinder has a volume of 0.04 m3 at 50 kg/cm2 and a volume of 0.039 m3 at 150 kg/cm2. The bulk modulus of elasticity of liquid is

(a) 400 kg/cm2 (b) 4000 kg/cm2 (c) 40 x 105 kg/cm2 (d) 40 x 106 kg/cm2 (e) none of the above. Ans: b

44. The units of viscosity are (a) metres2 per sec (b) kg sec/metre (c) newton-sec per metre2 (d) newtonsec per metre (e) none of the above. Ans: b

45. Kinematic viscosity is dependent upon (a) pressure (b) distance (c) level (d) flow (e) density. Ans: e

46. Units of surface tension are (a) energy/unit area (b) distance (c) both of the above (d) it has no units (e) none of the above. Ans: c

47. Which of the following meters is not associated with viscosity (a) Red wood (b) Say bolt (c) Engler (d) Orsat (e) none of the above. Ans: d

48. Choose the correct relationship (a) specific gravity = gravity x density (b) dynamic viscosity = kinematic viscosity x density (c) gravity = specific gravity x density (d) kinematic viscosity = dynamic viscosity x density (e) hydrostatic force = surface tension x gravity. Ans: b

49. Dimensions of surface tension are (a) MIL°T2 (b) MIL°Tx (c) MIL r2 (d) MIL2T2 (e) MIL°t. Ans:a

1 The capacity to do work is called as: A. Heat B. Energy C. work D. none of the above Ans: B

2 Heat is measured in: A. Joule B. Calorie C. both A and B D. Joule/second Ans: A It is measured in Joule .

3 1 cal. =? A. 1.2 joule B. 3.2 joule C. 4.2 joule D. none of the above Ans: C

4 The form of energy that produces feeling of hotness is called as: A. work B. Heat C. Energy D. none of the above Ans: B

5 With increase in temperature, heat will be: A. increase B. constant C. decrease D. double Ans: A Heat increase in temperature

7 The amount of heat required to raise temperature of a substance by 1°C is called as: A. work capacity B. heat capacity C. Energy capacity D. none of the above Ans: B

8 Heat capacity depends on A. change in temperature B. Mass of body C. Nature of substance D. All the above Ans: D It depends on (a) Mass of body (b) change in temperature (c) Nature of substance.

9 Heat bring …………….. change A. Physical B. chemical C. reversible D. periodic Ans: B Heat bring chemical change

10 …………………. is neither created nor destroyed it can only changed one form to another. A. work B. Heat C. Energy D. Mass of body Ans: C Energy is neither created nor destroyed it can only changed one form to another.

12 The amount of heat required to raise the temperature of 1 kg by 1°C is called as:

A. heat capacity B. work capacity C. specific heat capacity D. Energy capacity Ans: C

15 Which of the following are the processes of transfer of heat? A. Conduction B. Convection C. Radiation D. All the above Ans: D

16 The process of transfer of heat in solids is called as: A. Convection B. Radiation C. Conduction D. none of the above Ans: C In this process the molecules of the solid pass the heat from one to another, without themselves moving from their positions.

17 The temperature at which liquid changes into vapour is called as: A. Melting point B. boiling point. C. expansion point D. none of the above Ans: B

19 The process of transfer of heat in liquids & gases is called as: A. Conduction B. Radiation C. Convection D. absorption Ans: C It is the process of transfer of heat in liquids & gases

20 In convection, the molecules: A. without themselves moving from their positions. B. themselves move from one place to another C. without themselves moving from one place to another. D. None of the above Ans: B In convection, the molecules themselves move from one place to another, carrying heat with them.

22 It is the process of heat transfer from a hot body to a colder body without heating the space between the two is called as: A. Conduction B. Radiation C. Convection D. absorption Ans: B

23 At night a current of air blows from the colder land to the warmer sea is called as: A. air Breezes B. Sea Breezes C. Land Breeze D. none of the above Ans: C the sea is warmer than the land at night. So at night a current of air blows from the colder land to the warmer sea. This is called the land breeze.

24 The transfer of heat by radiation: A. does not require any medium. B. require any medium. C. does not require any space. D. require any space. Ans: A The transfer of heat by radiation does not require any medium.

29 A wooden spoon is dipped in a cup of ice cream. Its other end. A. becomes cold by the process of radiation. B. becomes cold by the process of conduction. C. "does not become cold." D. becomes cold by the process of convection. Ans: C

44 Radiation is the transfer of heat by means of: A. magnetic wave B. electromagnetic waves. C. electrical wave D. none of the above Ans: B

45 The effect of a material upon heat transfer rates is often expressed in terms of a number known as: A. Electrical conductivity B. conductivity C. thermal conductivity D. none of the above Ans: C The effect of a material upon heat transfer rates is often expressed in terms of a number known as the thermal conductivity.

56 If the area through which heat is transferred is increased by a factor of 2, then the rate of heat transfer is ……………….. . A. increased by a factor of 2 B. decreased by a factor of 2 C. increased by a factor of 4 D. decreased by a factor of 4 Ans: A

60 The equation relating the heat transfer rate to these variables is: A. Rate = k•(T1 – T2)/d B. Rate = k•A•(T1 – T2)/d C. Rate = A•(T1 – T2)/d D. Rate = k•A•(T1 – T2)/r Ans: B the transfer of heat through a glass window from the inside of a home with a temperature of T1 to the outside of a home with a temperature of T2. The window has a surface area A and a thickness d.

63 Kelvin scale is also called as: A. Celsius scale B. absolute scale C. Fahrenheit scale D. All the above Ans: B Kelvin scale is also called absolute scale of temperature.

66 The effects of heat on an object are : A. never change in shape of a body B. Change in temperature of a body C. never change state of matter D. All the above Ans: B Change in temperature of a body

1. A force of 50 N acts uniformly over and at right angles to a surface. When the area of the surface is 5 m2, the pressure on the area is: (a) 250 Pa (b) 10 Pa (c) 45 Pa (d) 55 Pa

2. Which of the following statements is false? The pressure at a given depth in a fluid

(a) is equal in all directions (b) is independent of the shape of the container (c) acts at right angles to the surface containing the fluid (d) depends on the area of the surface

3. A container holds water of density 1000 kg/m3. Taking the gravitational acceleration as 10 m/s2, the pressure at a depth of 100 mm is: (a) 1 kPa (b) 1 MPa (c) 100 Pa (d) 1 Pa

4. If the water in Question 3 is now replaced by a fluid having a density of 2000 kg/m3, the pressure at a depth of 100 mm is: (a) 2 kPa (b) 500 kPa (c) 200 Pa (d) 0.5 Pa

5. The gauge pressure of fluid in a pipe is 70 kPa and the atmospheric pressure is 100 kPa. The absolute pressure of the fluid in the pipe is: (a) 7 MPa (b) 30 kPa (c) 170 kPa (d) 10/7 kPa

6. A U -tube manometer contains mercury of density 13600 kg/m3. When the difference in the height of the mercury levels is 100 mm and taking the gravitational acceleration as 10 m/s2, the gauge pressure is:

(a) 13.6 Pa (b) 13.6 MPa (c) 13 710 Pa (d) 13.6 kPa

7. The mercury in the U-tube of Question 6 is to be replaced by water of density 1000 kg/m3. The height of the tube to contain the water for the same gauge pressure is:

(a) (1/13.6) of the original height (b) 13.6 times the original height (c) 13.6 m more than the original height (d) 13.6 m less than the original height

8. Which of the following devices does not measure pressure? (a) barometer (b) McLeod gauge (c) thermocouple

1. When a certain weight is suspended from a long uniform wire, its length increases by one cm. If the same weight is suspended from another wire of the same material and length but having a diameter half of the first one then the increase in length will be. A. 0.5 cm B. 4 cm C. 2 cm D. 8 cm

2. The bulk modulus of an ideal gas at constant temperature. A. Is equal to its volume V B. Is equal to its pressure p C. Is equal to p/2 D. Can not be determined

4. The area of cross-section of a wire of length 1.1 metre is 1 mm2. It is loaded with 1 kg. If Young's modulus of copper is 1.1 * 10 11 N/m 2 , then the increase in length will be (If g = 10). A. 0.01 mm B. 0.1 mm C. 0.075 mm D. 0. 15 mm

5. If Young's modulus of iron is 2*10 11 N/m 2 and the interatomic spacing between two molecules is 3*10 -10 metre, the interatomic force constant is. A. 60 N/m B. 30 N/m C. 120 N/m D. 180 N/m

7. A steel wire is stretched with a definite load. If the Young's modulus of the wire is Y. For decreasing the value of Y A. Radius is to be decreased B. Length is to be increased C. Radius is to be increased D. None of the above

8. The force constant of a wire does not depend on A. Nature of the material B. Length of the wire C. Radius of the wire D. None of the above

10. The ratio of the lengths of two wires A and B of same material is 1: 2 and the ratio of their diameter is 2: 1. They are stretched by the same force, then the ratio of increase in length will be. A. 2: 1 B.1: 8 C. 1: 4 D. 8: 1

12. The Young's modulus of a wire of length L and radius r is Y N/m2. If the length and radius are reduced to L/2 and r/2, then its Young's modulus will be. A. Y/2 B. 2Y C. Y D. 4Y

13. If the temperature increases, the modulus of elasticity A. Decreases B. Remains constant C. Increases D. Becomes zero

16. Hook's law defines. A. Stress B. Modulus of elasticity C. Strain D. Elastic limit

17. In a wire of length L, the increase in its length is l. If the length is reduced to half, the increase in its length will be A. l B. l/2 C. 2l D. None of the above

19. The extension of a wire by the application of load is 3 mm. The extension in a wire of the same material and length but half the radius by the same load is. A. 12 mm B. 15 mm C. 0.75 mm D. 6 mm

21. The compressibility of a material is A. Product of volume and its pressure B. The fractional change in volume per unit change in pressure C. The change in pressure per unit change in volume strain.